ActInf Textbook Group ~ Cohort 7 ~ Session 11 (Chapter 4) ~ 9/23/2024

TextbookGroup/ParrPezzuloFriston2022/Cohort_7 | Cohort 7 ~ Session 11 (Chapter 4) ~ 9/23/2024 | 1:02:00

okay welcome back cohort 7 we're
discussing chapter 4
so anyone can bring up any comment or
anything they remembered or a question
from
four or some other comment from earlier
on uh or we'll look at the chapter or
look at questions but first does anyone
have a comment or just any other
anything else they want to kick it off
with
okay type in the chat or or raise your
hand or
anything
um chapter 4 has two roads leading into
it with chapter 2 and
three so this is really where we see
active inference generative models after
the more General context of one and then
the low and the high road in in chapters
2 and
three
um anyone like is there a part of four
that anyone wants to go to or just a
quote that's interesting just to begin
okay
4.1 single
paragraph describes in very concise kind
of llm like style what the next sections
are going to
be uh section
4.2 goes
from base equation base theorem exact
base which is exactly how you update the
prior into the posterior given an

observation that's a calculation that is
done like in one Fell
Swoop the idea of using some bound on
exact basing inference so the idea of
doing approximate basing inference using
some other proxy not surprise exactly
but abound on surprise like variational
free energy the idea here is that that
bound could be picked to have certain
properties like be able to optimize it
incrementally with gradient descent like
in machine
learning like the title of the section
they do start with base theorem
though this chapter also says that it
will present the active inference
generative models and it does there are
a few kind of
background detours or or um
interludes like Jensen's
inequality towards equation
4.2 uh they use the definition of prize
as log likelihood and then through the
usage of that log or just any function
that has like a
decreasing uh rate
of decrease of the slope but never
turning back
negative um the log
surprisal allows like a um strict bound
to be described
which is
called the free energy or negative free
energy like just depending on the
negative
sign
um okay Jonathan
a previous uh participant and a

math teacher I think
Professor has written this
document let's
just see it for a second some of these
it's
like okay I think this date is
automatically chosen but that would be
interesting if he updated it so recently
let's look through
this while there are some superb
explanations in the book there are
important details left out in the
derivations which may not be that
important on the first pass but I think
make things much Clear when you want to
understand it in
detail our general end will be to
observe things we expect to observe we
will come on later to the things that we
want to observe erve when we think about
preferences but for now we can just
think about the things that a model
predicts that we are likely to
see here's the part with Jensen's
inequality so exact base 2.1 just like
in the book Jensen's
inequality this is going into more lines
of detail from what is in the book
this gives some
um lines to
follow this bottom line here here's
surprisal log likelihood this is the
likelihood of that
observation and then log likelihood for
for a variety of reasons to to get some
01 uh interval properties to get that
boundedness um to be able to compare
multiple uh like uh orders of magnitude

of
probability so this is whatever it
is here's the K
Divergence between like the distribution
that you can control your surprisal from
and uh
the empirical distribution
P um chapter 2.5 or chapter 2 equation
2.5 had the variational free energy and
there was multiple lines it seems like
he's going into more detail
here connecting how free energy
described in Chapter
2 and and then expected free energy
in equation
2.6 and then connecting it more to
chapter
4 so that's that's also kind of a
background math like a standalone math
thing any
ideas or questions on 4.1 or 4.2
so 4.1 again single paragraph yeah Jeff
go for it yeah no um I was recently
reading a paper and actually talking to
Hector Dean and Magdalena um
about um
how the decision making process you know
binary tree
branching it doesn't
scale it needs to be
bounded right like we can only make one
decision at one point in time but we're
Li limited by
our you know by the body that we inhabit
and our signal processing
when you H when you have systems of
systems who
are you know a scale-free right

uh
nested agents right who were
interdependent they they loss and
compression
there in order
to isolate the signal from the noise
and also you know have a pruning
function for that that that that
[Music]
um decision the the decision
tree that's just where my mind is right
now thanks Jeff yeah these are good um
you you point to these kind of
constraints on
the inbound with the signal
processing and then the the constraints
on the outbound I mean the ultimate
constraint on the
outbound one embodied body being able to
only do one like posture at once so it's
this tension between okay you could plan
out all these multi-joint
actions couple of joints a couple of
affordances a couple of time steps that
could be like all the atoms of the
universe and then
so there it's it's it's really
interesting because Seth Lloyd in
programming the universe which he wrote
back in 2006 he shows very clearly that
like you know a a game board not big not
much bigger than a than a go set has
more potential States than all the all
the matter and energy in the
universe right so like you know in one
run of the universe you will not explore
all the potential states that could
possibly exist

yeah and I I I don't know about the
branching uh like Multiverse but
sometimes it's like that's just like
saying it's kind of acceptance of
that whether whether as a uh whether
that's an epistemically strong stance or
whether that's a a cope but it's just
like saying yeah that much branching
really does happen it's like okay but it
doesn't really get around
that like chessboard
explosion and then I mean that just is
essentially an
empirical validation that cognitive
heris discs are used for sense making
and decision
making yeah even like when you model
like very small sequential
domains um it becomes very unwieldly
very quick as you scale the board
one one actin uh connection there is
this uh branching time active inference
Direction and maybe a little bit of more
recent work this is this is uh drawing
methods from uh dealing with these tree
and combinatoric explosions whether
that's whether you think about that in
the combinator explosions of
interpretation on the inbound or like
action s joint action planning uh or
time or plan that kind of tree search
has been a classic problem like chess
algorithms and and deep blue and Beyond
so yeah and alphago and Alpha star and
whatnot right they' they've solved these
for a specific
domain
um exactly and and

broadly some of those approaches are
Monte Carlo based Monte Carlo is like a
city where gambling happens so these are
the gambling sampling based
approaches and then an alternative
complimentary approach like some
problems with it both neither one or the
other do well Etc but another approach
is to cast that as a variational
inference or as a basing inference um
problem and then use very ational
methods to approximate the
posterior whereas Monte Carlo samples
and and Lance daosta has done a lot of
work on the sampling and the
variational uh
approaches but that that's
yeah a few other random thoughts like
it's it's easy to make a a cognitive
behavioral model that's like it's just
like maps that don't to territory as
well for for one reason another mean you
could make it wildly overpowered like
you could have it be a a mouse foraging
model with a gigabyte of short-term
memory of the GPS location it's like the
model won't
complain that mice don't actually
remember the GPS
coordinates so it um H taking into
account those
constraints is often like the hard or a
hard part of empirical modeling whereas
if one is just more interested in like a
looser like bio inspired approach and
then is seeking performance or some
other attribute just Loosely based upon
biology then of

course there there's there's nothing
holding back but to actually connect and
make useful predictions for a given like
biological system then how how um that
gets represented becomes challenging
because it isn't just about making like
the highest
performance model
because it's more about fitting to
empirical
data there's many ways in practice to to
deal with this
ranging from core
screening
data to picking
simpler variational
distributions using uh different kinds
of model architectures so that there
there aren't like wide computations like
um the all by all by all like you just
figure out other
ways to structure the problem so that
the um computational proper ities are
not so
disadvantageous if if it really even is
a problem at that scale
um there can be major heris STS like um
if the input data were 4K video I mean
one approach you could take that in as a
huge Matrix with like millions of
numbers and then in that case maybe it
is hard to run on but then maybe um
another approach like uh passes the
video frame to an llm and then gets back
estimate on which items are in it and
then takes in that list of
items so there's a lot of ways to to
deal with the and and and also just to

study okay how does the runtime change
as a function of like the size of the
input or the number of sensors like is
it linear as adding more sensors or
sublinear or is it super
linear 4.2 is is a another background
but a really important intuition figure
this sets us up to look at the base
graphs that will be like the PDP and
represent the perception and action
components but um
happily this is is not
unlike a uh systems dynamic or systems
modeling or just kind of like mind map
or flowchart
representation I mean um if you just
talking about day-to-day
situations like there was a lightning
strike and then there was light and
sound I mean maybe somebody wouldn't
draw the two and the three boxes just
draw X directly connected to Y and to W
but
topologically this is pretty
much what is is conversationally
described when people are talking about
um observed and unobserved
phenomena any random thoughts on that or
or like just things people seeing this
or about these base graphs overall I saw
the muszle muzzle flash and then I saw
the
impact similar yeah here that's like
this
one
yeah that same idea that okay nodes are
going to represent
distributions being between zero and one

like probability distributions um or
just random variables like it could be
number between one and 10 or could be um
just different kind different kinds of
uh support for those variables different
values that they could take on uh so
it's kind of just like a variable in a
program um that same idea that variables
are declared and then their their
connections and influences are
represented with edges and some of these
are observables the O's meaning that
they're kind of
Taken even if they're understood to be a
noisy sensor still the observation is
taken as it
is um and then
unobserved V variables like the latent
States and then the the sort of
connection between signal processing on
bottom and control theory on the top and
then that work is done with the B Matrix
three which is the probability function
that maps The Hidden State like the real
temperature in the room at the next time
Point conditioned upon the current time
point and the current behavior so this
is a uh action
conditioned Markov transition
Matrix it s is like a vector it could be
like one temperature data point in the
room or it could be like a vector of of
thermometers this B is like a stack of
index cards each card is s by S
Dimensions it just describes from and
two so if you don't have any affordances
or if there's just one affordance like
this the the just this downstairs part

of the system is unfolding with with no
action selection then it's just like
from and two and that's like a Markov
trans position Matrix and then policy
selection is just selecting which of
those index cards is going to be used
for picking uh which the transition to
the next time point and then kind of a a
simpler but analogous role is played by
the a
matrix which can be used to go from a
given hidden state to a distribution
over outcomes so it's like I know that
my thermometer has plus or minus one you
know uh measurement noise so if it were
really 81 then I would expect this
distribution between 80 and
82 that's using it in the forward
Direction like generating synthetic data
and then also there's taking in a point
measurement and then updating your
prior to what your best belief is now H
not shown here but that's basically
where attention comes into
play if you are not attending to the
observation then it has no impact if you
have complete attention to the
observation then it overrules your prior
and you update your belief to the
observation so that's like the Precision
or the kind of um sharpness on a
and and and how it relates to the
strength of the
product figure
4.3 uh shows on the top the discrete
time setting and on the bottom The
Continuous time
setting the the kind of

uh structural similarity is highlighted
like with using the same box numbering
for the action selection
State transition and the sense making
component one one three and two
boxes at the same time different
notation letters are used like o for the
observations indry time and Y um for
continuous
time highlighting structural
similarity with generative models
whether they used screet time or
continuous time while also highlighting
their differences and the major
difference is that in discrete time
discreet specific time steps are
considered like s of t t plus1 t plus2
for a given
Horizon whereas in the continuous time
setting Which is less about like
transitions between states discrete
States and more about like a continuous
differentiable line
passing through the present this is more
analogous to the Taylor series
expansion so there are not actually
future time points specifically
predicted the only specific uh
evaluation of the function is at the the
current time point and then the past and
the future get fit with a tailor series
being able to fit further and further
out with more and more
approximations so plus here for for for
uh one low cost you if you do have a
prediction for every past and future
data
point the downside is it's hard to even

know how
relevant that prediction is so like here
we take the first uh we take the first
pass which is just evaluating it at the
current moment that's just this point
then you take the first derivative that
point and you get the red line so it's
like okay so it's going up in the in the
right and it's going down the left but
then let's say you expand your
approximation and now you're in the
third uh
powering but e even though in the area
around zero it still has the same slope
but now there's a totally different
story with it going off to Infinity
quickly to the left and then down to the
right and then go into the fourth
powering
and and it's maybe it's a different
story again like with coming so even
though again it's all being guaranteed
to to get more and more accurate in the
neighborhood
but but uh
knowing even getting a Precision on well
how close is my prediction like to the
real one that can be challenging to know
so then it is an empirical question like
how good of approximation is this Taylor
series
expansion this gets Revisited in chapter
seven and eight so I mean this is kind
of the main generative model description
of the book since two and three get us
to the generative
models and six is about how
to design them and five is about some

specific ones that are used and then
nine about getting connected with the
data this figure and then chapter seven
and 8 which it sort of
anticipates are the
main generative model
descriptions so this is like a micro
7 yeah go for col yeah so that that
applies also to the action of doing
nothing right so
on the different potential policies one
is always doing nothing right so that if
I do nothing for instance it doesn't
matter if it's in the continuous or or
the discrete
approximation
um um you you you can refine your
inference either by making continuous
observations by doing nothing even if
you don't interact with the system right
or or the system itself could could
without doing anything from your side
could evolve and they naturally evolve
so that applies also to that policy to
the policy of doing nothing right is
correct yeah great great comment like um
H how is doing nothing dealt
with so sometimes let's let's just say
we're in a uh grid World movement
sometimes do nothing is is incorporated
as an afford
so there's like up down left right
stay another approach would be um higher
higher
level stay or or
move and then lower
level up down left
right

so those either those could be used um
and I mean you you point to a really
important question which is the control
theory and Game Theory it's often just
framed like it's obvious when action
should be taken and so in a turn-taking
game like
chess you you could talk about like well
you know when should this Pawn you know
advance or something like that but
that's that's within the game time um
it's a it's an alternating turn taking
but then when there's a system that's
changing
external then like I mean like a an
economic Market it's like timing the
market there there might be way broader
changes that your actions are less
influential on such that for for um for
achieving like pragmatic or epistemic
value it it almost matters less which
action is done but more about like when
done so those that's like kind of
another flavor of of control
theoretic question but it's not really
so easy to address in the turn-taking
game
setting but it is more kind of amenable
to to ecosystems where things are really
happening yeah more Dynamic way uh ones
in which you know you you cannot infer
the state just by one observations you
need to wait for a number ofs they come
along in order to be able to make a good
inference for of of the current
state yeah I mean that's that's even
another angle which is like let's just
say that um measurements cost us one

like there's just a cost per
measurement um so we have a budget
so we can't just you know get a million
measurements in in a split second but
then also there's like a rate of change
of the underlying system this is like
buying stock market trading data or like
planning and experimental data so then
there it's this assessment of like well
for deciding on what to do that's like
the actionable side what is the tradeoff
of like which measurements and when to
make those
measurements yeah
and yeah some of these are are would be
really interesting to kind of to kind of
sketch out and just show how different
heris sixs would lead to like
different kind of
um biases in the sense making and
decision making like if you just if you
in if you if you have a a c you have a
certain number
of credits for how much measurements you
can make well if you immed immediately
pursue the highest
Precision then you're initially even if
it is initially accurate then that
assessment would become
inaccurate if you wait until the very
end you cross the finish line like with
the highest accuracy but then for for
most of the time you had the least of an
idea well if if you space it out then
like you may have a lower level but but
above
nothing so then like that kind of comes
back to this constraints

topic and and how the interesting pieces
about how do you how do you model those
constraints because that's where the the
decision like the kind of wide open
space that's why it's so hard to predict
what an
eoli I mean or a mouse will
do yep yeah but but yeah definitely
about doing nothing is as that's
that like or or I'm just thinking of in
in like an agent based model sometimes
there even is no Stay
option so then there's like a lot of
movement just moving between two squares
back and forth like if they're already
on kind of a local Maxima or something
but it's like but that that kind of
behavior isn't really
seen but that just because there isn't
the Stay
option but just like walking around like
the bird is just like sitting on the
tree not like spending not resting by
hopping back and
forth but that's like a that's a that's
a huge difference but that's just about
having that's just about the structure
of the model and how it's being
fit yeah than yeah also it highlights
how like free energy is very model
specific so just whether thinking of the
perceptual variational free energy
calculation like just the sense making
from noisy and partial data or the
action selection
as an
inference uh
problem that free energy

calculation is based upon the
data the family of variational
distributions the generative
model so then that it it yes that's an
instrumentalist perspective but it it
highlights why like well what's the free
energy of that squirrel why is it
sitting on the ground like that it's
like you're quite a ways off but if you
described okay the data or the
squirrel's location the variational
distributions of the gine that I'm
making and the prior and the likelihood
on location for the
squirrel here's why it is or isn't
surprising to me why my model is
returning a high or a low free energy
for the squirrel being
there but that's more of just describing
your statistical model
but but just a a a sort of
qualitative description of free energy
as if it was just naturally an
unambiguous part of a of a biological
systems kind of a
stretch I'm reminded of the halting
problem like
how like you can you know write endless
loops
kind of kind of reminds me of uh you
know hoffstead or two you know good aler
and Bach and uh uh I am a strange
Loop while at uh I'm gonna welcome back
to
the this was
uh uh in this this Workshop over the
previous days Chris Fields gave us talk
I'll put it in our in our uh page and

I'll put it in the
chat and it just sort of there was it
was just some interesting discussions
like well you know never step in the
same river twice we're we're never the
same we're we're we're whether you take
that from a material like breathing CO_2
and oxygen um food um or just
information and like planning and stuff
like that it's like we're never the same
so
then
what this part was really
interesting the neural like the I don't
know if this comes from the icade or
from the motor like analogy function or
something like that
but the embodied
math and and just the the way that the
category the and I think other equations
get
um
interpreted then it's like so then what
is the same it's like well the identity
is what is the sameness but then it's
like but then the but then your identity
When You're 15 when you're 30 it's that
is
different so it's just like I I I I I
don't think his his
DOA answers it per se but it gives some
some interesting stick drawings to
begin exploring like that sameness and
differentness which is one of the most
kind of
pervasive
crossmodal aspects of experience yet at
the same time it's it's it's really not

so simple when the our identity
materially and cognitively and
metacognitively is changing but it does
have a sameness too
and he mentions octm and uh connects a
little bit to the some he doesn't
imagine it but but the the the
um physics is information
processing but just showing like the
sort of uh um usage of a of a very
ational free energy or something like
that for like a phenomenological human
State this is just prompted by the the
geser bach it's
like the whole point is that's a strange
system it can't just be taken for
granted like even what direction what
layer of
what it that's why there isn't like an
open source kernel for for the human
mind it just
it it it it seems to be so
baffling and
strange again revisiting equation 2.5
and
2.6 variational free
energy F of Q and
Y it's with a belief distribution and a
new data point so it's just like a real
time
um real time through time
minimization it can include it can be a
sense making only or it can include like
instantaneous action inference so
without explicit
Planet single time step updating of
beliefs including beliefs about action
whereas if you want to get into explicit

planning than expected free energy is
used this is a lot like equation
2.6 still in the discreet time section
this is just going through more details
of stuff that's going to come back in
chapter 7 4.5 continuous
box a blanketed off
box on marov
blanket um with its
definition Upstream causes of
X Downstream effects of X and the
co-parents of the children of
X after that short
definition immediately going into these
two methods kind of like Network
propagation or percolation methods for
having messages passed among nodes that
implement this
variational uh gradient
descent here the uh it's shown with kind
of another notation
set um
the left is showing a a sort of
classical predictive processing
hierarchical model so observations
through time are coming in O
Tilda each of these four that look
closer together are one layer of the
hierarchical predictive processing like
hierarchy um at each level
there's um a descending
prediction that's getting
contrasted with the with the ascending
observation and then that is getting
like turned into an error residual it's
getting passed back up to a higher
level um so here's like primary sensory
processing

level
um another nested level and then I think
just the error unit for like an overall
so this is like measurement coming in
this is like the ones place this is like
the 10's place and then this is just
like green light yellow light or red
light how accurate is the 10's Place
prediction that's so that's a snapshot
or or kind of uh Through Time
emphasizing the hierarchical predictive
architecture whether or how the brain
does it is secondary to here it's just
this is a modeling Motif that's used in
the basian
systems here on the
right um is looking more through time so
here just like in the top of figure 4.3
it's observations coming at the current
time point and the past and the present
uh past in the future time point and
then just showing a little bit of like
kind of the information flow of how
those updates get
pasted um
in order to update beliefs about
policy including like and updating
hidden
States conditioned on
policies it it it says it emits some
intermediate connection so it's
like not exactly sure
um how much to look into the edge
connections but
that the it is fully described but here
they they don't show all the
edges here now we're kind of back into
this this continuous time case this is a

really important set of equations the
SPM statistical parametric mapping or
like anything involving
neuroimaging this is kind of a classic
pair as well as in physics so
neuroimaging this is the observations
coming in with like a signal and a noise
could be a sensor noise and then x dot
is like a
spatialized flow landscape of an
underlying State like neural activity so
the the measurement of fmri might be
like the um just the intensity of the
signal coming from a given voxal at a
time that's the
observation which which is due to it to
underlying like blood flow changes and
also noise and then there's this
underlying neural activity landscape
which is x
dot so here's a situation where they're
both normal uh distributions
4.2 generalized coordinates of motion is
the tailor
series you can take the tailor series or
generalized coordin motions on the
observation and on the latent State x
dot so the these are derivatives first
second Etc
derivatives and and and again in the
kind of classic hierarchical predictive
processing scheme that the the Precision
comes into
play llao approximation free energy may
be approximated by a quadratic expansion
around the posterior mode so the mode is
the most common like the highest point
on the probability

distribution the median is the 50th
percentile data point the mean is like
average value the mode is just picking
the highest point which is like the
single maximum likelihood estimate and
then the llao approximation is taking um
to the second power around that point so
you have a quadratic
curvature and then fitting that quad
ratic
distribution so if if if it's not a if
there's not a central tendency then of
course this will still only be able to
fit a quadratic so if it's like three
humps then you're either going to have a
really over dispersed quadratic or have
one that just fits over one but um it's
an
approximation now it's again the
hierarchical and the through time but
instead of the O's it's the Y's so now
it's kind of swapped into the notation
set of the continuous
time that's chapter 4
five is gonna gonna do some reviews
on neural systems which is where these
models have been applied most but yeah
four is where the the generative
models at least the simple motifs for
discret and continuous time are
shown and there there's there's other
motifs and patterns like hierarchical
and things like that but um
they highlight the difference and
similarities between the continuous and
discreet time and that's like a big
structuring element of the book with
chapter 7 and 8

Daniel just a question uh I haven't seen
yet any uh real world problem that has
been using this type of hierarchical
models everything I've seen so far is
flat can you give me some examples of
references links to
uh works that use hierarchical model
please yeah good good question for for
this robot navigation
paper describes um okay it's not an Open
Access
paper here's their hierarchical model so
this is one
setting in a multiscale movement like
locations in a warehouse and then robot
body postures that's that's like one
where the higher and the lower level are
both um like mechanical and
positional
um the hierarchical hybrid model with
continuous on the lower level and
discret on the top is kind of the folk
psychologic iCal
template that's that's that um in this
Smith at all paper I mean they argue
this is basically the um implicit or
explicit generative model and people
talk about in a folk psychological
sense let's see if they have
any I don't know if they have a
visualization but um
and then I think um but okay that those
are a
few um in the implementations I agree
they're usually higher they're usually
single um
layer
uh it could be a range of things ranging

from Advanced models um happening uh
proprietarily another option is that a
model like um for example Alex
Aria's work and uh these kind of the
neural uh generative coding this this
does make hierarchical models so there's
a flatness because it's like a neural
topology a neuromorphic system but then
the calculation it implements could
still be
hierarchical and I think another um
interesting direction to
explore djar hner dream to control
Information Gain in the planning
objective Carl first mentioned sometimes
dreamer is like um some variation of
active inference
so I I think the
textbook clear
clearly shows the model doesn't have to
be hierarchical like architecturally or
functionally for it to be active
inference even for for perhaps
interesting or multiscale behavior to
arise definitely showing more clear
nested models though will be interesting
and understanding like okay if you have
to do 100 time steps let's see the 100
times step planner the 10 times step
planner the decade and then the the ones
place planner the no
planner the single times like kind of
create in all of those alternative ways
to to parse the
problem I think would be very
informative because there's probably
niches and inputs and and uh sense
making and action problems for which

like but there's no free lunch but but
on on some data set each of those you
could imagine could
be effective or
not but for real world settings it's
definitely an empirical question like
how because if if it's just about like
well the world the systems are
hierarchical or they're nested but then
that's just getting into the the map
being the
territory I'm has anybody ever framed it
in such a way
of the so so like reinforcement learning
and deep learning right we're getting
these massive data sets and we're trying
to figure out how to make an agent move
about the world and it feels to me like
the framing of act active inference as a
way for an agent to move the world
around
it almost like comes to
mind so instead of moving moving around
a bounded environment right the the the
agent is like moving the whole like the
agent is the
Central folky of
existence I I don't know what's been
done but sometimes that's called like
egocentric
navigation where the ref point is
whereas like in the grid world examples
that we look at in pmdp the agent like
Square two comma two but it's like but
how' you know you were in a 3X3 maze
whereas for an open space than than the
egocentric navigation and then it also
connects like to this um to the the

action and non-action earlier because
it's like um I mean one nistic of
movement and and plausibly something
that is done for movement or like being
in the car it's like well you could
think about the that the world and and
its causes are fixed and then you're
moving over that or it's like being on a
treadmill
it's like yeah the freeway is like
passing
forward so then from the egocentric
frame that's why it feels stationary
even like being on a
train but then from from an allocentric
frame you might think about yourself
moving positionally across a fixed
space so and again that goes to like
making these animal behavioral models
that are just like wildly off
base like people will for example set up
an L-shaped wall and then they'll test
okay well do do animals have an internal
space representation like will they take
the hypotenuse cuz it's shorter to get
to the food or or will they just follow
the edge but it's like but the animal
may not even have the vision to see
across the diagonal or they may just
have a a habit to stay up against a wall
because it's
safer so it's like
you but so then having made a made a um
allocentric model where it's like okay
it's going to calculate a hypotenuse and
then it's going to minimize the path
distance it's like it's just has nothing
to do with the her istics of the

animal so then then then what whatever
the P value you get there is it's just
like it's a model output but it it
doesn't help address like an interesting
question about the animal in in such a
clean
way you know I'd be it'd be really
interesting to explore this in terms of
the Interior changes that
occur um in the predictive models of of
animals that undergo
metamorphosis like um you know a
caterpillar to a butterfly like or um
the uh spawn of um Barnacles right like
they they start as a U mobile little and
they can swim about the ocean and then
they fix onto something and then like
they're fixed and the world then is
anchored for
them yeah also like the I mean the
multi-generation like monarch
butterfly are they using a are they just
you could imagine all of the you could
put a GPS tracker Etc think of them
moving through space or it could be a
pure
heris based upon like
light and magnetism and seasonality all
these
things that doesn't require planning but
then back again back to this
combinatorics of planning it's like well
if it's combinatorics of every muscle
twitch it's like well if if that's how
if if yeah that's maybe not what is
happening but this is this is the other
map territory
what and the Butterfly is able to do

that but but can't NE necessarily
emulate chess so even just getting the
trick of detering complete computer to
get constraints and capacities that even
resemble the biological system is
totally not a
given and the and the model won't throw
up a flag again if you give it a you
give the butterfly a GPS coordinate and
and and a Bluetooth communication with
the rest it's like you can make that
model but it doesn't have the biological
plausibility
okay thank you see you next time
thanks bye byebye